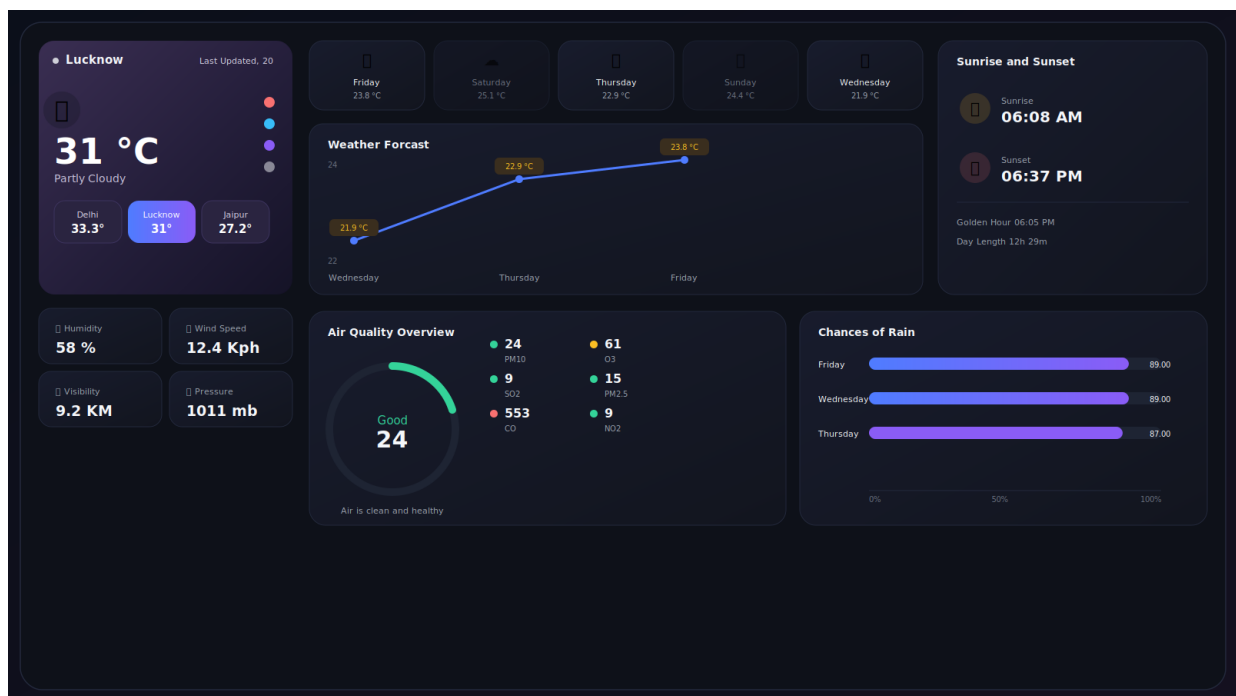


Weather Analytics Dashboard Pro

Project Documentation



Microsoft Power BI • Power Query • DAX • Dark Glassmorphism UI
10 Cities • 365 Days • 6 Report Pages • 40+ DAX Measures

1. Project Overview

Weather Analytics Dashboard Pro is a production-ready Power BI project covering current conditions, 7-day forecasts, air quality, and historical trends for ten major Indian cities. The design language is dark glassmorphism: translucent rounded cards, soft drop shadows, blue-to-purple gradients, and Segoe UI typography in the style of Microsoft's Fluent UI.

Component	Detail
Platform	Microsoft Power BI Desktop
Data volume	10 cities x 365 days (3,650 weather + 3,650 AQI rows)
Forecast horizon	7 days per city
Report pages	Overview, Forecast, Air Quality, Historical Trends, City Comparison, Weather Alerts
Theme	Dark background, blue (#4F7CFF) + purple (#8A5CF6) accents, rounded 16px corners
Measures	40+ DAX measures across temperature, atmosphere, forecast, trend, AQI, and KPI-color groups

2. Installation

- 1 Install Power BI Desktop (free from Microsoft Store or powerbi.microsoft.com).
- 2 Get Data -> Text/CSV and import City.csv, Calendar.csv, Weather.csv, Forecast.csv, AQI.csv from /Data.
- 3 In Power Query Editor, open the Advanced Editor for each query and paste the matching M script from PowerQuery/Queries.txt.
- 4 Close and Apply.
- 5 Create a _Measures table and add every DAX measure from DAX/Measures.txt.
- 6 Build the star-schema relationships shown in Section 3.
- 7 View -> Themes -> Browse for themes -> select Theme/theme.json.
- 8 Lay out each report page following Section 5 and the reference mockup on the cover.
- 9 Save As Weather Dashboard.pbix.

3. Data Model

The model follows a star schema: two dimension tables (City, Calendar) and three fact tables (Weather, Forecast, AirQuality), each joined to City and Calendar on a many-to-one relationship. Keeping every fact table on the many side avoids bidirectional filtering and keeps the model performant even as the fact tables grow.

Table	Type	Grain	Key Columns
City	Dimension	1 row / city	CityID, City Name, Region, Lat/Long
Calendar	Dimension	1 row / day	Date, Month, Quarter, Season, IsWeekend
Weather	Fact	1 row / city / day	Temperature, Humidity, Wind, Pressure, Condition
Forecast	Fact	1 row / city / forecast day (7)	MaxTemp, MinTemp, RainProbability, Sunrise/Sunset
AirQuality	Fact	1 row / city / day	AQI, PM2.5, PM10, CO, NO2, SO2, O3

4. Power Query

All transformations live in PowerQuery/Queries.txt as ready-to-paste M scripts, including null handling, explicit type correction, column renaming, a merge example (Weather joined to City for fast tooltips), an append example (Weather actuals combined with Forecast into a single continuous timeline), calculated columns, and an optional live WeatherAPI.com variant using Web.Contents.

5. DAX Measures

DAX/Measures.txt contains 40+ measures organized into eight groups: core temperature, atmospheric conditions, forecast, trend/rolling averages, air quality, dynamic KPI color measures (hex strings for conditional formatting), a composite Weather Score, and utility measures for the unit toggle and last-refresh label.

Measure	Purpose
Current Temperature	Latest daily temperature reading in context
Rolling Average (7D)	7-day moving average temperature for trend lines
Trend %	Week-over-week percentage change in average temperature
Weather Score	0-100 composite comfort index (temp, humidity, AQI, rain, visibility)
AQI Color	Hex color string driving conditional formatting on AQI cards

6. Dashboard Layout

The layout follows an icon-first weather-app reference style: a compact forecast strip, a ring gauge with colored-dot pollutant tiles, and horizontal probability bars, rather than a generic grid of charts.

Left panel — Purple-to-dark gradient card: location, last-updated time, condition icon, large current temperature, and three horizontal city-selector chips.

Below left panel — 2x2 grid of Humidity, Wind Speed, Visibility, and Pressure cards.

Top strip — Five compact forecast chips (icon, day name, temperature), alternating opacity to highlight the active days.

Center — Weather Forecast line chart with floating temperature-value pills above each point and day names along the bottom.

Top right — Sunrise and Sunset card with icon rows, plus Golden Hour and Day Length below a divider.

Bottom center — Air Quality Overview: ring gauge (AQI + category label) alongside six pollutant tiles (PM10, PM2.5, O3, NO2, SO2, CO), each with a colored status dot, plus a one-line summary caption.

Bottom right — Chances of Rain: horizontal gradient bars per day with percentage labels and a 0/50/100% axis.

Additional Pages

Forecast (full 7-day + hourly detail), Air Quality (city-by-city matrix and trend), Historical Trends (365-day line/area charts with rolling averages), City Comparison (small multiples across all 10 cities), and Weather Alerts (threshold-driven table, e.g. AQI above 200 or rain probability above 70%).

7. Customization

- Colors: edit Theme/theme.json (dataColors array, background/foreground values).
- Add a city: append a row to City.csv plus matching rows in Weather.csv, Forecast.csv, AQI.csv.
- Live data: swap in the WeatherAPI.com Power Query variant at the bottom of Queries.txt.
- Unit toggle: use the bundled Display Temperature / Temp Unit Toggle Label measures with a disconnected parameter table.

8. Deployment

- Power BI Service: Home -> Publish to a workspace; schedule refresh (gateway needed only if data stays on-premises).
- Power BI Embedded: embed the published report via the powerbi-client SDK on Premium/Embedded capacity.
- Template distribution: File -> Export -> Power BI Template (.pbit) to share the design without the sample data.

9. Future Improvements

- Real-time streaming dataset for live current-conditions cards.
- Incremental refresh policy on the Weather and AirQuality fact tables.
- Custom radar-overlay visual on the Map page.
- Anomaly-detection analytics pane on Historical Trends.
- Power BI mobile layout view.
- Row-level security by Region for multi-tenant use.

Note: all data in /Data is synthetically generated for demonstration and does not reflect real observed weather. See README.md for the full build guide.